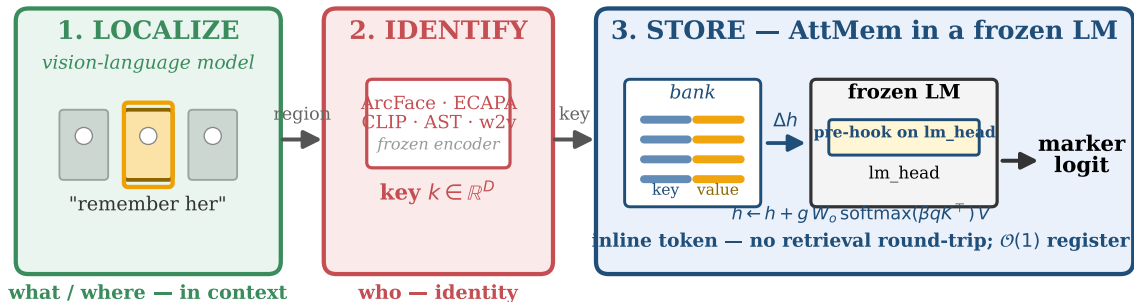


Factored perceptual memory: localize (VLM) → identify (encoder) → store (in-model token)



VLM alone: weak identity (0.54). Encoder alone: cannot localize (0.05). Factored: recovers the oracle (0.96).